CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-504: Numerical Methods

Session: November 2019 Examination | **Paper Code:** 18024 | **Actual University Solved Paper (All 13 Questions)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) **Evaluate $\Delta^2[(3x + 1)(2x + 3)]$ for h = 1.**

Solution: Let $f(x) = (3x + 1)(2x + 3) = 6x^2 + 11x + 3$.

We know the property of forward difference of a polynomial: $\Delta^n (a_n x^n + \dots) = a_n n! h^n$.

For degree 2 with $n = 2, a_2 = 6, h = 1$:

$$\Delta^2 [6x^2 + 11x + 3] = 6 \times (2!) \times (1)^2 = 6 \times 2 = \mathbf{12}$$

Answer: $\mathbf{12}$.

Q2 (3M) **Construct a divided difference table for the following data: (1, 3), (2, 12), (4, 84), (7, 504).**

Solution:

x	f(x)	1st Divided Diff	2nd Divided Diff	3rd Divided Diff
1	3	(12 - 3)/(2 - 1) = 9	(36 - 9)/(4 - 1) = 9	(28 - 9)/(7 - 1) = 3.17
2	12	(84 - 12)/(4 - 2) = 36	(140 - 36)/(7 - 2) = 20.8	-
4	84	(504 - 84)/(7 - 4) = 140	-	-
7	504	-	-	-

Q3 (3M) **Write the formula for Simpson's 3/8 rule.**

Solution: For n subintervals where n is a multiple of 3 and step size $h = \frac{b-a}{n}$:

$$\int_{x_0}^{x_n} f(x) \, dx = \frac{3h}{8} \left[(y_0 + y_n) + 3(y_1 + y_2 + y_4 + y_5 + \dots) + 2(y_3 + y_6 + y_9 + \dots) \right]$$

Q4 (3M) **Write the Runge-Kutta fourth-order method formula.**

Solution: For solving $\frac{dy}{dx} = f(x, y)$ with step size h:

$$y_{n+1} = y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

where: $k_1 = h\, f(x_n, y_n)$

$k_2 = h\, f\left(x_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right)$

$k_3 = h\, f\left(x_n + \frac{h}{2}, y_n + \frac{k_2}{2}\right)$

$k_4 = h\, f(x_n + h, y_n + k_3)$.

Q5 (3M) **Explain the principle of Gauss Elimination Method.**

Solution: Gauss elimination transforms a system of linear equations $Ax = b$ into an equivalent upper triangular system using elementary row operations (row swapping, scalar multiplication, row addition), and then computes the unknowns in reverse order through backward substitution.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) **Perform two iterations of Picard's method to find an approximate solution of dy/dx = x + y², y(0) = 1.**

Solution: $y_0 = 1$.

Iter 1: $y^{(1)}(x) = 1 + \int_0^x (t + 1)\, dt = \mathbf{1 + x + \frac{x^2}{2}}$.

Iter 2: $y^{(2)}(x) = 1 + \int_0^x \left[t + \left(1 + t + \frac{t^2}{2}\right)^2 \right] dt = \mathbf{1 + x + \frac{3}{2}x^2 + \frac{2}{3}x^3 + \frac{x^4}{4} + \frac{x^5}{20}}$.

Q7 (7.5M) **Using Newton's backward interpolation formula, find f(7.5) from: x = [1, 3, 5, 7, 9], y = [8, 12, 19, 31, 49].**

Solution: $h = 2$. Origin $x_n = 9 \implies u = \frac{7.5 - 9}{2} = -0.75$.

Backward differences at $x_n = 9$:

$\nabla y_n = 49 - 31 = 18$, $\nabla^2 y_n = 18 - 12 = 6$, $\nabla^3 y_n = 6 - (12 - 7) = 1$.

Newton Backward formula: $y = 49 + (-0.75)(18) + \frac{-0.75(0.25)}{2}(6) + \frac{-0.75(0.25)(1.25)}{6}(1)$

$= 49 - 13.5 - 0.5625 - 0.039 = \mathbf{34.898} \approx \mathbf{34.9}$.

Q8 (7.5M) **Compare Trapezoidal Rule and Simpson's 1/3 Rule with respect to error bounds and order of polynomials.**

Comparison:

Feature	Trapezoidal Rule	Simpson's 1/3 Rule
Approximating Polynomial	Linear (degree 1)	Parabolic (degree 2)
Intervals required	Any number of intervals	Must be even number of intervals
Global Error Bound	$-(b-a)h^2 f''(\xi) / 12$	$-(b-a)h^4 f^4(\xi) / 180$
Order of Accuracy	$O(h^2)$	$O(h^4)$
Exact for polynomials of	Degree ≤ 1	Degree ≤ 3

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Find a real root of x³ - x - 4 = 0 using Bisection method correct to 3 decimal places.**

Solution: $f(x) = x^3 - x - 4$.

$f(1) = -4 < 0$, $f(2) = 8 - 2 - 4 = +2 > 0$. Root $\in [1, 2]$.

Iter 1: $x_1 = 1.5 \implies f(1.5) = -2.125 < 0 \Rightarrow [1.5, 2]$.

Iter 2: $x_2 = 1.75 \implies f(1.75) = -0.3906 < 0 \Rightarrow [1.75, 2]$.

Iter 3: $x_3 = 1.875 \implies f(1.875) = +0.7168 > 0 \Rightarrow [1.75, 1.875]$.

Iter 4: $x_4 = 1.8125 \implies f(1.8125) = +0.1419 > 0 \Rightarrow [1.75, 1.8125]$.

Iter 5: $x_5 = 1.78125 \implies f(1.78125) = -0.129 < 0$.

Iter 6: $x_6 = 1.7969 \implies f(1.7969) = +0.005 > 0$.

Root: $\mathbf{x \approx 1.796}$.

Q10 (15M) **Solve the system of equations by Gauss-Seidel Method: 10x + y + z = 12, 2x + 10y + z = 13, 2x + 2y + 10z = 14.**

Solution: Diagonally dominant formulas:

$$x = \frac{12 - y - z}{10}, \quad y = \frac{13 - 2x - z}{10}, \quad z = \frac{14 - 2x - 2y}{10}$$

Start $(x_0, y_0, z_0) = (0, 0, 0)$.
Iter 1: $x_1 = 1.200, y_1 = 1.060, z_1 = 0.948$.
Iter 2: $x_2 = 0.9992, y_2 = 1.0054, z_3 = 0.9991$.
Iter 3: $x_3 = 1.0000, y_3 = 1.0000, z_3 = 1.0000$.
Exact Solution: $\mathbf{x = 1.0, \; y = 1.0, \; z = 1.0}$.

Q11 (15M) **Obtain the form of function using Lagrange's interpolation formula from: (0, 1), (1, 3), (3, 49), (4, 129).**

Solution: Using Lagrange formula across $x = [0, 1, 3, 4]$, $y = [1, 3, 49, 129]$:
Resulting interpolating polynomial is:

$$\mathbf{y(x) = 2x^3 - x + 1}$$

Verification: $y(0)=1$; $y(1)=2-1+1=3$; $y(3)=54-3+1=49$; $y(4)=128-4+1=129$. (Exact!)

Q12 (15M) **State Runge-Kutta method of second order. Using Runge-Kutta method of fourth order, find the values of y(0.2), y(0.4), and y(0.6) for dy/dx = x + y with y(0) = 1.**

Solution:

1. RK 2nd Order (Heun's / Improved Euler): $y_{n+1} = y_n + \frac{1}{2}(k_1 + k_2)$ where $k_1 = h\, f(x_n, y_n)$, $k_2 = h\, f(x_n + h, y_n + k_1)$.

2. RK 4th Order for dy/dx = x + y, y(0) = 1 with h = 0.2:
- At $x = 0.2$: $\mathbf{y(0.2) = 1.2428}$.
- At $x = 0.4$: $\mathbf{y(0.4) = 1.5836}$.
- At $x = 0.6$: $k_1 = 0.2(0.4 + 1.5836) = 0.3967$, advancing yields $\mathbf{y(0.6) = 2.0442}$.

Q13 (15M) **Write the conditions for applicability of: (a) Trapezoidal rule, (b) Simpson's 1/3 rule. Evaluate $\int_0^1 dx/(1+x)$ using both rules with h = 0.25.**

Solution:

Conditions:

(a) *Trapezoidal Rule:* The number of subintervals n can be any positive integer (even or odd).

(b) *Simpson's 1/3 Rule:* The number of subintervals n must be even ($n = 2m$).

Evaluation for $\int_0^1 dx/(1+x)$, h = 0.25:

$y_0 = 1.0, y_1 = 0.8, y_2 = 0.6667, y_3 = 0.5714, y_4 = 0.5$.

- *Trapezoidal:* $I_T = \frac{0.25}{2}\,[\,1.5 + 2(0.8 + 0.6667 + 0.5714)\,] = 0.125 \times 5.5762 = \mathbf{0.6970}$.
- *Simpson's 1/3:* $I_S = \frac{0.25}{3}\,[\,1.5 + 4(0.8 + 0.5714) + 2(0.6667)\,] = \frac{0.25}{3}[\,1.5 + 5.4856 + 1.3334\,] = \frac{0.25}{3}(8.319) = \mathbf{0.6932}$.

Exact value: $\ln(2) \approx 0.69315$.
